

CureAdapter: Efficient Test-Time Adaptation of Large Language Models Towards Therapeutic Decision Making

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Challenges: Medical and Therapeutic Domain Adaptations

White-box LLMs. (1) A significant *performance discrepancy* still exists between larger black-box LLMs (e.g., GPT-3.5-Turbo) and smaller white-box LLMs (e.g., LLaMAs), even when the latter are fine-tuned on biomedical-specific corpora. (2) Moreover, fine-tuning even a moderately-sized LLM with 7B parameters requires *substantial computational resources*, often exceeding the capabilities of many academic and medical centers.

Black-box LLMs. (1) Uploading patient data via APIs poses significant risks of *privacy leakage* and potential conflicts with HIPAA compliance, including unauthorized third-party access to personally identifiable information (PII); (2) Fine-tuning API services could incur prohibitively *high financial and environmental costs*, exceeding typical academic or clinical budgets; (3) The opaque fine-tuning process, limited to very few adjustable hyperparameters within a specific range, often results in *suboptimal performance* in downstream tasks, whereas drug applications often demand precise outcomes.

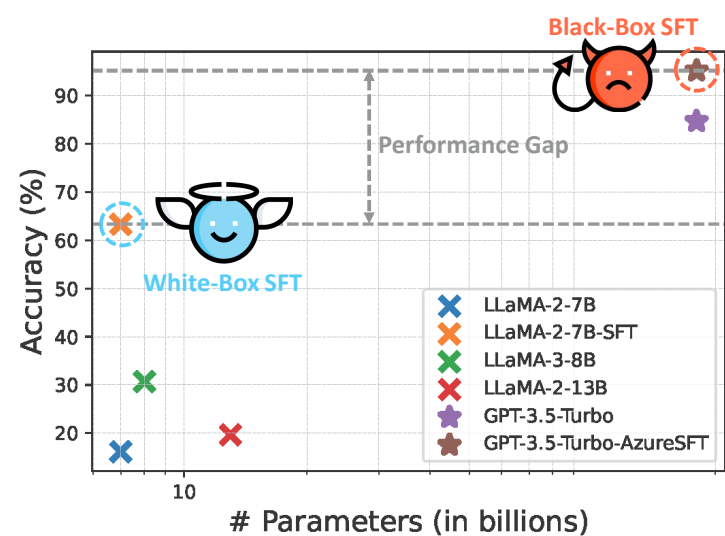


Figure 1. Preliminary evaluation results on a biomedical QA task, BioASQ.

CureAdapter: Adapting LLMs to Therapeutic Reasoning

We introduce CureAdapter, a unified test-time adapter that fine-tunes a **lightweight language model** (even as small as BERT-size, 110M) to facilitate the adaptation of both white-box and black-box LLMs for therapeutic reasoning.

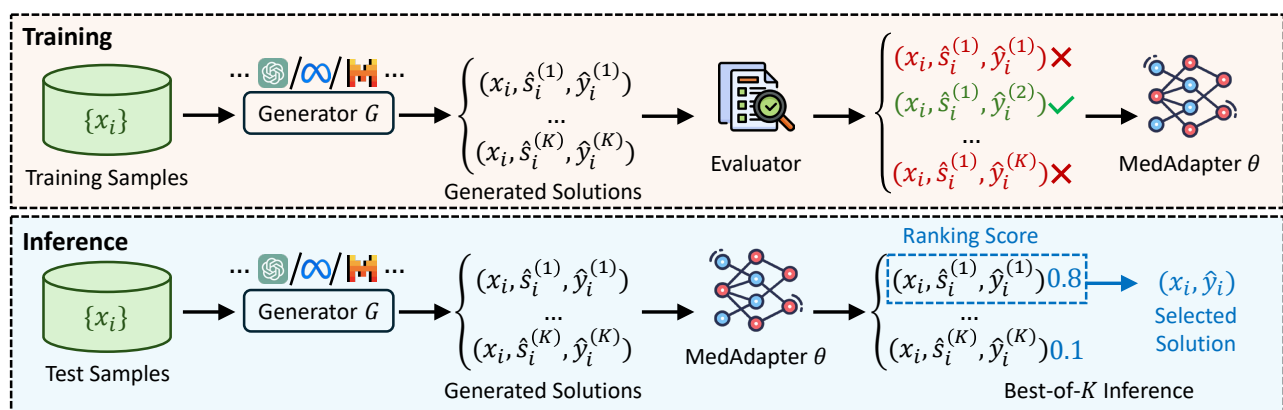


Figure 2. Overview of CureAdapter for efficient test-time LLM adaptation towards therapeutic reasoning. We fine-tune a small adapter to rank candidate solutions generated by LLMs, thereby effectively establishing a distinction between the source and target domains for efficient domain adaptation.

- **Candidate Solutions Generation.** For each problem $\mathbf{x}_i$ in the training dataset $\mathcal{D}$, we generate k intermediate candidate reasoning solutions $\{\hat{\mathbf{s}}_{i,j}\}_{j=1}^k$ (e.g., chain-of-thought rationales or multi-step reasonings) and the corresponding answer $\{\hat{\mathbf{y}}_{i,j}\}_{j=1}^k$ using greedy decoding with the language model generator G . With access to ground-truth answers $\mathbf{y}_i$, we can verify the correctness of each generated solution $\hat{\mathbf{y}}_{i,j}$ and assign a corresponding binary correctness label z_i as:

$$z_i = \mathbb{1}(\hat{\mathbf{y}}_{i,j} = \mathbf{y}_i), z_i \in \{0, 1\}. \quad (1)$$

- **Outcome-Supervised Adapter.** More specifically, given a therapeutic reasoning problem $\mathbf{x}$ and its corresponding candidate solutions $\hat{\mathbf{y}}$, the outcome verifier ($V: \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$) assigns a normalized adaptation score, ranging from 0 to 1, to each solution to indicate the correctness.
- **Best-of-K Inference.** During the inference stage, for each test question $\mathbf{x}_i$, we adopt the best-of- K inference, often referred to as rejection sampling, to select the best solution from multiple candidates. We initially sample K candidate solutions $\{\hat{\mathbf{s}}_{i,j}, \hat{\mathbf{y}}_{i,j}\}_{j=1}^K$ from the generator G . The solution with the highest adaptation score is then selected:

$$\hat{\mathbf{y}}_i = \arg \max_{j=1, \dots, K} r_\theta([\mathbf{x}_i || \hat{\mathbf{s}}_{i,j} || \hat{\mathbf{y}}_{i,j}]). \quad (2)$$

- **Implementation Details.** In the preliminary explorations, we employ **LongFormer-Base** (110M) as the base language model for CureAdapter. We set $k = 8$ for all generations of intermediate candidate reasoning solutions using CureAdapter.

Remark: CureAdapter as Efficient Test-Time Adapter

In contrast to prior verification-guided in-context learning methods that depend on large-scale intermediate reasoning annotations, CureAdapter utilizes candidate solutions generated by LLMs to form positive and negative examples, thus **removing the need for human-annotated intermediate reasoning steps**. Additionally, the lightweight design of the adapter results in a **minimal increase in memory usage and inference time**. In the final evaluation, we achieve the final score of 0.79097 on the testing set.

Extensive Experimental Results on Biomedical QA

Due to the lack of annotations on the final testing set in CureBench, we conducted extensive experiments on additional **four biomedical reasoning tasks across 8 datasets** demonstrate that CureAdapter effectively adapts both white- and black-box LLMs for medical reasoning, achieving average performance improvements of 18.24% and 10.96%.

Dataset (→)	MedMCQA		MedQA		MMLU-Med		PubMedQA		BioASQ		MedNLI		MedQA-RQE		PubHealth	
Method (↓) / Metrics (→)	Acc. (%)	Δ (%)	Acc. (%)	Δ (%)	Acc. (%)	Δ (%)	Acc. (%)	Δ (%)	Acc. (%)	Δ (%)	Acc. (%)	Δ (%)	Acc. (%)	Δ (%)	Acc. (%)	Δ (%)
LLaMA-2-7B	16.00	–	16.42	–	20.13	–	17.00	–	16.13	–	17.80	–	23.91	–	16.89	–
+Self-Consistency	21.20	+5.20	22.39	+5.97	23.27	+3.14	28.00	+11.00	17.74	+1.61	27.87	+10.07	25.22	+1.31	17.79	+0.90
+CureAdapter	32.00	+16.00	32.52	+16.10	27.67	+7.54	58.00	+41.00	62.90	+46.77	30.46	+12.66	27.39	+3.48	19.25	+2.36
+SFT†	42.86	+26.86	33.39	+16.97	28.22	+8.09	60.80	+43.80	63.31	+47.18	65.52	+47.72	35.42	+11.51	22.00	+5.11
BioMistral-7B	28.95	–	29.77	–	33.33	–	26.20	–	28.53	–	22.03	–	42.37	–	25.73	–
+Self-Consistency	29.18	+0.23	32.68	+2.91	39.62	+6.29	30.60	+4.40	31.45	+2.92	31.46	+9.43	44.68	+2.31	28.90	+2.17
+CureAdapter	30.31	+1.36	34.88	+5.11	46.54	+13.21	33.20	+7.00	33.06	+4.53	35.96	+13.93	45.53	+3.18	30.84	+5.11
LLaMA-3-8B	20.44	–	27.81	–	25.16	–	19.00	–	30.65	–	21.96	–	49.13	–	27.70	–
+Self-Consistency	26.87	+6.43	31.50	+3.69	31.45	+6.29	37.00	+18.00	33.06	+2.41	30.12	+8.16	50.87	+1.74	35.01	+7.31
+CureAdapter	32.08	+11.64	32.44	+4.63	35.22	+10.06	55.00	+36.00	64.52	+31.46	32.09	+10.13	51.74	+2.61	36.07	+8.37
LLaMA-2-13B	19.66	–	28.04	–	22.01	–	47.40	–	19.66	–	21.75	–	30.44	–	19.33	–
+Self-Consistency	28.40	+8.74	31.03	+2.99	28.30	+6.29	56.80	+9.40	51.61	+31.95	24.21	+2.46	43.04	+12.60	24.70	+5.37
+CureAdapter	32.00	+12.34	37.47	+9.43	33.96	+11.95	63.60	+16.20	65.32	+45.66	26.88	+5.13	44.78	+14.34	27.46	+8.13
gpt-3.5-turbo	49.74	–	61.51	–	59.75	–	56.00	–	84.68	–	66.64	–	50.00	–	23.38	–
+Self-Consistency	56.20	+6.46	67.71	+6.20	69.81	+10.06	71.60	+15.60	87.90	+3.22	69.18	+2.54	51.30	+1.30	25.41	+2.03
+MedRAG	51.80	+2.06	64.36	+2.85	68.85	+9.10	50.00	–6.00	87.55	+2.87	–	–	–	–	–	–
+CureAdapter	59.02	+9.28	68.66	+7.15	73.58	+13.83	73.40	+17.40	93.55	+8.87	75.09	+8.45	52.61	+2.61	33.43	+10.05
+Azure-SFT†	61.82	+12.08	63.32	+1.81	70.55	+10.80	71.40	+15.40	95.16	+10.48	91.27	+24.63	58.08	+8.08	36.56	+13.18
gpt-4	69.48	–	83.90	–	85.53	–	69.20	–	92.74	–	86.77	–	51.30	–	38.52	–
+Self-Consistency	70.08	+0.60	84.05	+0.15	86.79	+1.26	72.20	+3.00	93.54	+0.8	87.26	+0.49	51.74	+0.44	43.35	+4.83
+MedRAG	66.65	-2.83	82.80	-1.10	87.24	+1.71	70.60	-1.40	92.56	-0.18	–	–	–	–	–	–
+CureAdapter	72.09	+2.61	84.13	+0.23	87.42	+1.89	77.40	+8.20	95.97	+3.23	87.68	+0.91	53.04	+1.74	46.34	+7.82

Table 1. Main results (accuracy) of adapting white-box and black-box LLMs to biomedical tasks. † denotes the upper bound in theory using supervised fine-tuning (SFT).

Complementary Analysis. When combined with train-time adaptation, CureAdapter outperforms either train-time or test-time adaptation alone, demonstrating its utility as a flexible solution that complements existing train-time adaptation methods (e.g., LoRA) and even test-time adaptation (e.g., MedRAG) to further boost model performance.

Dataset (→)	MedMCQA		MedQA		MMLU-Med		PubMedQA		BioASQ		MedNLI		MedQA-RQE		PubHealth	
Method (↓) / Metrics (→)	Acc. (%)	Δ (%)	Acc. (%)	Δ (%)	Acc. (%)	Δ (%)	Acc. (%)	Δ (%)	Acc. (%)	Δ (%)	Acc. (%)	Δ (%)	Acc. (%)	Δ (%)	Acc. (%)	Δ (%)
LLaMA-2-7B	16.00	–	16.42	–	20.13	–	17.00	–	16.13	–	17.80	–	23.91	–	16.89	–
+CureAdapter	32.00	+16.00	32.52	+16.10	27.67	+7.54	58.00	+41.00	62.90	+46.77	27.87	+10.07	25.22	+1.31	17.79	+0.90
+SFT	42.86	–	33.39	–	28.22	–	60.80	–	63.31	–	65.52	–	35.42	–	22.00	–
+CureAdapter	44.85	+1.99	40.61	+7.22	35.85	+7.63	68.00	+7.20	66.94	+3.63	74.95	+9.43	46.54	+11.12	33.36	+11.36
+SFT-LoRA	28.95	–	23.89	–	24.54	–	55.00	–	50.00	–	25.97	–	32.17	–	20.47	–
+CureAdapter	35.69	+6.74	28.04	+4.15	31.90	+7.36	62.90	+7.90	60.48	+10.48	35.96	+9.99	39.57	+7.40	25.26	+4.79
gpt-3.5-turbo	49.74	–	61.51	–	59.75	–	56.00	–	84.68	–	66.64	–	50.00	–	23.38	–
+CureAdapter	59.02	+9.28	68.66	+7.15	73.58	+13.83	73.40	+17.40	93.55	+8.87	75.09	+8.45	52.61	+2.61	33.43	+10.05
+Azure-SFT	61.82	–	63.32	–	70.55	–	71.40	–	95.16	–	91.27	–	58.08	–	36.56	–
+CureAdapter	65.50	+3.68	68.89	+5.57	76.73	+6.18	77.00	+5.60	95.97	+0.81	91.42	+0.15	59.56	+1.48	42.49	+5.93
+MedRAG	51.80	–	64.36	–	68.85	–	50.00	–	87.55	–	–	–	–	–	–	–
+CureAdapter	56.20	+4.40	67.16	+2.80	74.86	+6.01	63.00	+13.00	94.42	+6.87	–	–	–	–	–	–

Table 2. Complementary analysis results (accuracy) of combining training- and test-time adaptation for both white- and black-box LLMs on biomedical tasks.

Cost Analysis. For black-box LLMs, it achieves comparable performance or even surpasses fine-tuning APIs at only 15.59% of the budget, while also eliminating the risks associated with private data sharing.

Dataset (→)	MedMCQA		MedQA		MMLU-Med		PubMedQA		BioASQ	
Method (↓) / Costs (\$)	Training	Inference	Training	Inference	Training	Inference	Training	Inference	Training	Inference
gpt-3.5-turbo	–	1.37	–	0.67	–	0.06	–	0.16	–	0.03
+CureAdapter	7.67	10.40	42.57	5.37	3.49	0.44	0.92	1.14	1.41	0.35
+Azure-SFT	71.18	10.88	172.85	6.83	38.93	3.18	38.17	3.76	38.48	3.24
+OpenAI-SFT*	23.07	32.87	195.45	16.10	4.01	3.12	15.76	1.34	6.77	1.05

Table 3. Cost (\$) estimations of adapting black-box LLMs to biomedical tasks based on gpt-35-turbo.

Parameter Efficiency. CureAdapter reaches 99.35% of supervised fine-tuning performance using only 14.75% of the GPU memory on BioASQ. Notably, CureAdapter demonstrates its efficiency by achieving 87.50% of the performance of the fully supervised fine-tuning model while only updating an adapter comprising 110M parameters, which constitutes **merely 1.57% of the parameters (7B) of the original model**.

Dataset (→)	BioASQ		
Method (↓) / Memory (GiB)	Training	Inference	Acc. (%)
LLaMA-2-7B	–	25.42	16.13
+CureAdapter	11.60	33.00	62.90
+SFT-LoRA	54.76	34.65	50.00
+SFT	78.65	25.42	63.31

Loss (↓) / Dataset (→)	BioASQ	MMLU	MedMCQA
InfoNCE	87.90	69.18	57.43
Pairwise	92.74	72.33	59.83
Cross-entropy (Ours)	93.55	73.58	59.02

Table 4. GPU memory (GiB) usage estimations of adapting white-box LLMs to biomedical QA tasks.

Scale-up Analysis. CureAdapter serves as a scoring function that heavily relies on language comprehension rather than generative capabilities, which is a natural fit to encoder-only model. The limited fine-tuning data available may allow smaller models to more effectively capture underlying patterns within the candidate solutions.

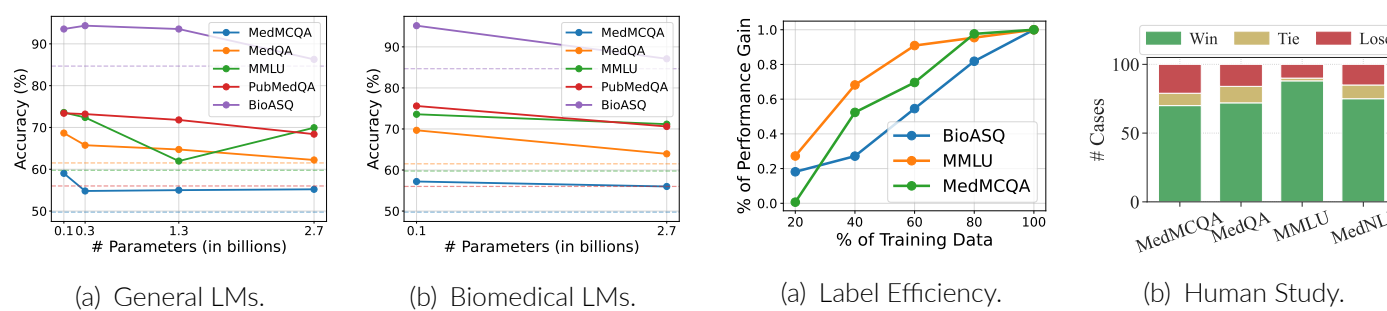


Figure 3. Scale-up performance.

Figure 4. Empirical and Human Analysis.